

AC circuits - cheatsheet

1. AC Wave Equation

$$[v = V_m \sin(\omega t)]$$

$$[i = I_m \sin(\omega t)]$$

Where:

- (V_m) = Maximum (peak) voltage
 - (I_m) = Maximum (peak) current
 - $(\omega = 2\pi f)$
-

2. Frequency and Time Period

$$[f = (1/T)]$$

$$[T = (1/f)]$$

$$[\omega = 2\pi f]$$

3. RMS Values

$$[V_{rms} = (V_m)/(\sqrt{2})]$$

$$[I_{rms} = (I_m)/(\sqrt{2})]$$

4. Average Value (Half Cycle)

$$[V_{avg} = 0.637 V_m]$$

$$[I_{av} = 0.637 I_m]$$

Average over a full cycle = 0

5. Reactance

Inductive Reactance

$$[X_L = 2\pi fL]$$

Capacitive Reactance

$$[X_C = (1/(2\pi fC))]$$

6. Impedance

Pure Resistance

$$[Z = R]$$

RL Circuit

$$[Z = \sqrt{R^2 + X_L^2}]$$

RC Circuit

$$[Z = \sqrt{R^2 + X_C^2}]$$

RLC Circuit

$$[Z = \sqrt{R^2 + (X_L - X_C)^2}]$$

7. Current

$$[I = (V/Z)]$$

8. Voltage Across Elements

$$[V_R = IR]$$

$$[V_L = IXL]$$

$$[V_C = IXC]$$

9. Phase Angle

RL Circuit

$$[\tan(\phi) = (XL/R)]$$

RC Circuit

$$[\tan(\phi) = (XC/R)]$$

RLC Circuit

$$[\tan(\phi) = (XL - XC)/R]$$

10. Power Factor

$$[\cos(\phi) = (R/Z)]$$

- RL → Lagging

- RC → Leading
 - RLC → Depends on (XL) and (XC)
-

11. Power

Active (Real) Power

$$[P = VI \cos \phi]$$

Unit = Watt (W)

Reactive Power

$$[Q = VI \sin \phi]$$

Unit = VAR

Apparent Power

$$[S = VI]$$

Unit = VA

12. Power Triangle

$$[S^2 = P^2 + Q^2]$$

13. Resonance

At resonance,

$$[X_L = X_C]$$

$$[Z = R]$$

$$[I = (V/R)]$$

Current is **maximum**.

Power factor = **1 (Unity)**

14. Three-Phase Formulae

Star (Y)

Phase Voltage

$$[V_{ph} = (V_L / (\sqrt{3}))]$$

Line Voltage

$$[V_L = \sqrt{3} V_{ph}]$$

Current

$$[I_L = I_{ph}]$$

Delta (Δ)

Voltage

$$[V_L = V_{ph}]$$

Current

$$[I_L = \sqrt{3} I_{ph}]$$

Three-Phase Power

$$[P = \sqrt{3} V_{LL} I_L \cos \phi]$$

15. Inductance and Capacitance

From reactance,

$$[L = (X_L / (2\pi f))]$$

$$[C = (1 / (2\pi f X_C))]$$

DC generator - transformer : -----

DC Generator EMF

$$[E_g = ((P \phi Z N) / 60 A)]$$

- Lap winding: $(A = P)$
- Wave winding: $(A = 2)$

Shunt Generator

- $(I_f = (V / R_f))$
- $(I_a = I_L + I_f)$
- $(E_g = V + I_a R_a)$

Transformer Efficiency

- Iron loss = OC test power
- Full-load copper loss = SC test power

- Copper loss at (x) per unit load:

$$[P_{Cu} = x^2 P_{Cu}(FL)]$$

- Efficiency:

$$[\eta = (\text{Output} / (\text{Output} + \text{Losses})) \times 100\%]$$

Induction Motor : -----

1. Synchronous Speed

$$[N_s = (120f/P)]$$

Where:

- (N_s) = Synchronous speed (rpm)
- (f) = Frequency (Hz)
- (P) = Number of poles

2. Slip (Decimal)

$$[S = ((N_s - N) / (N_s))]$$

Where:

- (S) = Slip
- (N) = Rotor speed (rpm)

3. Percentage Slip

$$[\%S = ((N_s - N) / (N_s)) \times 100]$$

4. Rotor Speed

$$[N = N_s(1-S)]$$

Note: Here, (S) must be in **decimal form**, not percentage.

Example:

- Slip = 5% = 0.05
-

5. Slip Speed

$$[\text{Slip Speed} = N_s - N]$$

6. Rotor Frequency

$$[f_r = S \times f]$$

Where:

- (f_r) = Rotor current frequency (Hz)
-

7. Rotor EMF at Slip (S)

$$[E_{2s} = S \times E_2]$$

Where:

- (E_2) = Rotor EMF at standstill
 - (E_{2s}) = Rotor EMF at slip (S)
-

8. Rotor Reactance at Slip (S)

$$[X_{2s} = S \times X_2]$$

Where:

- (X_2) = Rotor reactance at standstill
-

Energy Consumption : ----- -----

1. Power Conversion

$$[1 \text{ kW} = 1000 \text{ W}]$$

$$[\text{Power (kW)} = (\text{Power (W)})/1000]$$

2. Electrical Energy

$$[E = P \times t]$$

where:

- (E) = Energy (kWh)
 - (P) = Power (kW)
 - (t) = Time (hours)
-

3. For Multiple Appliances

$$[P_{\text{total}} = n \times P]$$

where:

- (n) = Number of appliances
 - (P) = Power of one appliance
-

4. Daily Energy Consumption

$$[E_{day} = P_{total} \times \text{Hours per day}]$$

5. Monthly Energy Consumption

$$[E_{month} = E_{day} \times \text{Number of days}]$$

or directly,

$$[E_{month} = ((n \times P(W) \times h \times d) / 1000)]$$

where:

- (n) = Number of appliances
 - (P) = Power (W)
 - (h) = Hours/day
 - (d) = Number of days
-

6. Total Energy

$$[E_{total} = E_1 + E_2 + E_3 + \dots]$$

7. Energy Charges

$$[\text{Energy Charges} = \text{Total Units} \times \text{Rate per Unit}]$$

8. Total Electricity Bill

[Total Bill = Energy Charges + Fixed Charges]

9. Unit of Electricity

[1 kWh = 1 Unit]

Exam Shortcut Formula

For any appliance:

[Units = $((n \times P(W) \times h \times d) / 1000)$]

Then,

[Bill = $(\text{Units} \times \text{Rate}) + \text{Fixed Charges}$]

These formulas are sufficient to solve almost all **energy consumption and monthly electricity bill** numerical problems in Basic Electrical Engineering (BEE).

===== problems

=====

These are standard **AC Circuits (RL, RC, RLC, and 3-phase)** problems. I'll explain every step, the formulas, and the solutions.

1. Current equation

Given

[$i = 77\sin(314t)$ A]

Compare with

$$[i = I_m \sin(\omega t)]$$

where

- (I_m) = Maximum current
- $(\omega = 2\pi f)$

Step 1: Maximum current

$$[I_m = 77 \text{ A}]$$

Step 2: Frequency

$$[\omega = 314]$$

$$[2\pi f = 314]$$

$$[f = (314/2\pi)]$$

$$[f = 50 \text{ Hz}]$$

Answer

- Maximum current = **77 A**
 - Frequency = **50 Hz**
-

2. RL Series Circuit

Given

$$[R = 10\Omega, X_L = 15\Omega]$$

Power factor

$$[\cos(\phi) = (R/Z)]$$

First calculate impedance

$$[Z = \sqrt{R^2 + X_L^2}]$$

$$[= \sqrt{10^2 + 15^2}]$$

$$[=\sqrt{325}]$$

$$[=18.03\Omega]$$

Power factor

$$[=(10/(18.03))]$$

$$[=0.555]$$

Since it is RL,

Power factor = 0.555 lagging

3. Series RLC Circuit

Given

- $R = 10\Omega$
- $L = 0.2 \text{ H}$
- $C = 100 \mu\text{F}$
- $V = 230\text{V}$
- $f = 50\text{Hz}$

Step 1

$$[XL=2\pi fL]$$

$$[=2\pi(50)(0.2)]$$

$$[=62.83\Omega]$$

Step 2

$$[XC=(1/(2\pi fC))]$$

$$[=(1/(2\pi(50)(100\times 10^{-6})))]$$

$$[=31.83\Omega]$$

Step 3

Net reactance

$$[X=X_L-X_C]$$

$$[=62.83-31.83=31\Omega]$$

Step 4

Impedance

$$[Z=\sqrt{(10^2+31^2)}]$$

$$[=32.57\Omega]$$

Step 5

Current

$$[I=(230/(32.57))]$$

$$[=7.06A]$$

Step 6

Power factor

$$[=(10/(32.57))]$$

$$[=0.307]$$

Since inductive,

Power factor = 0.307 lagging

4. RLC Circuit at Two Frequencies

Given

$$R=10\Omega$$

$$L=0.05H$$

$$C=300\mu F$$

$$V=100V$$

(a) $f=25Hz$

Inductive reactance

$$[X_L=2\pi(25)(0.05)]$$

$$[=7.85\Omega]$$

Capacitive reactance

$$[X_C=(1/(2\pi(25)(300\times 10^{-6})))]$$

$$[=21.22\Omega]$$

Net reactance

$$[X=7.85-21.22=-13.37\Omega]$$

Negative means capacitive.

Impedance

$$[Z=\sqrt{10^2+13.37^2}]$$

$$[=16.69\Omega]$$

Current

$$[I = (100 / (16.69))]$$

$$[= 5.99 \text{ A}]$$

Phase angle

$$[\varphi = \tan^{-1}((-13.37)/10)]$$

$$[= -53.2^\circ]$$

(Current leads voltage.)

(b) $f = 50 \text{ Hz}$

$$[X_L = 15.71 \Omega]$$

$$[X_C = 10.61 \Omega]$$

Net

$$[X = 5.1 \Omega]$$

Impedance

$$[Z = 11.23 \Omega]$$

Current

$$[I = 8.91 \text{ A}]$$

Phase angle

$$[= 27^\circ]$$

(Current lags.)

5. Series RLC

Given

$$R = 12 \Omega$$

$$L = 0.15 \text{ H}$$

$$C=100\mu\text{F}$$

$$V=100\text{V}$$

$$f=50\text{Hz}$$

Reactances

$$[X_L=47.12\Omega]$$

$$[X_C=31.83\Omega]$$

Net

$$[X=15.29\Omega]$$

Impedance

$$[Z=\sqrt{(12^2+15.29^2)}]$$

$$[=19.44\Omega]$$

Current

$$[I=(100/(19.44))]$$

$$[=5.14\text{A}]$$

Power factor

$$[=(12/(19.44))]$$

$$[=0.617]$$

Lagging.

Voltage drops

$$[V_R=IR]$$

$$[=61.7\text{V}]$$

$$[V_L=IX_L]$$

$$[=242\text{V}]$$

$$[V_C=IX_C]$$

$$[=164\text{V}]$$

Voltage phasor:

- $(V_R) \rightarrow$ horizontal
 - $(V_L) \rightarrow$ upward $(+90^\circ)$
 - $(V_C) \rightarrow$ downward (-90°)
 - Supply voltage is the vector sum of (V_R) and $((V_L - V_C))$.
-

Theory: Behaviour of Series RL Circuit

In an RL circuit:

- Resistor consumes real power.
- Inductor stores magnetic energy.
- Current lags voltage.
- Phase angle

$$[\tan(\phi) = (X_L/R)]$$

Power factor

$$[=\cos(\phi) = (R/Z)]$$

As frequency increases:

- (X_L) increases.
 - Current decreases.
 - Lag increases.
-

6. RL Circuit

Given

$$R = 150\Omega$$

$$L = 0.2H$$

$$V = 200V$$

Assume 50Hz.

Inductive reactance

$$[X_L = 62.83\Omega]$$

Impedance

$$[Z = \sqrt{(150^2 + 62.83^2)}]$$

$$[= 162.6\Omega]$$

Current

$$[I = (200 / (162.6))]$$

$$[= 1.23A]$$

Power factor

$$[= (150 / (162.6))]$$

$$[= 0.923]$$

Lagging.

Voltages

$$[V_R = 184.5V]$$

$$[V_L = 77.3V]$$

Active power

$$[P = VI \cos\phi]$$

$$[= 227W]$$

Reactive power

$$[Q = VI \sin\phi]$$

$$[=95\text{VA}_r]$$

7. Three-Phase Star Load

Given

$$[Z=15+j20]$$

Magnitude

$$[|Z|=25\Omega]$$

Power factor

$$[=(15/25)]$$

$$[=0.6]$$

Lagging.

Line voltage

$$[400\text{V}]$$

Phase voltage

$$[V_{ph}=(400/(\sqrt{3}))]$$

$$[=231\text{V}]$$

Phase current

$$[=(231/25)]$$

$$[=9.24\text{A}]$$

Since star,

Line current

$$[=9.24\text{A}]$$

Power

$$[P=\sqrt{3}V_{LL}I_L\cos\phi]$$

$$[=1.732\times 400\times 9.24\times 0.6]$$

$$[=3.84\text{kW}]$$

8. Pure Resistance

Given

$$[V=10\sin 314t]$$

Resistance

$$10\Omega$$

Peak voltage

$$[V_m=10\text{V}]$$

Peak current

$$[=(10/10)]$$

$$[=1\text{A}]$$

RMS current

$$[=(1/(\sqrt{2}))]$$

$$[=0.707\text{A}]$$

Average current (over one full cycle)

$$[=0\text{A}]$$

Time period

$$[T=(1/50)]$$

$$[=0.02\text{s}]$$

Frequency

$$50\text{Hz}$$

At

$$[t=10\text{ms}]$$

$$[V=10\sin(314\times 0.01)]$$

$$[= 10 \sin \pi]$$

$$[= 0V]$$

9. Series RLC

Given

$$R = 10\Omega$$

$$[X_L = 12\Omega]$$

$$[X_C = 18\Omega]$$

Current

$$10A$$

Voltage drops

$$[V_R = IR]$$

$$[= 100V]$$

$$[V_L = 120V]$$

$$[V_C = 180V]$$

Inductance

$$[L = (X_L / (2\pi f))]$$

$$[= (12 / 314)]$$

$$[= 0.0382H]$$

Capacitance

$$[C = (1 / (2\pi fXC))]$$

$$[= (1 / (314 \times 18))]$$

$$[= 177 \mu F]$$

Power factor

Net reactance

$$[X = 12 - 18 = -6\Omega]$$

Impedance

$$[Z = \sqrt{(10^2 + 6^2)}]$$

$$[= 11.66\Omega]$$

Power factor

$$[= (10 / (11.66))]$$

$$[= 0.857]$$

Since $(XC > XL)$, the circuit is **capacitive**, so the power factor is **0.857 leading**.

These are standard **DC Generator** and **Transformer** numerical problems. I'll teach the concept first, then solve each question step by step.

Question 1

An 8-pole, lap-wound armature has 1200 conductors and a flux per pole of 0.02 Wb. Determine the generated EMF when running at 600 rpm.

Formula

For a DC generator,

$$[E_g = \frac{P \Phi n Z N}{60 A}]$$

where

- (P) = number of poles = 8
- (Φ) = flux per pole = 0.02 Wb
- (Z) = total conductors = 1200
- (N) = speed = 600 rpm
- (A) = parallel paths

For **lap winding**

$$[A = P]$$

So,

$$[A = 8]$$

Substitute

$$[E_g = \frac{(8 \times 0.02 \times 1200 \times 600)}{(60 \times 8)}]$$

The 8 cancels.

$$[= \frac{(0.02 \times 1200 \times 600)}{60}]$$

$$[= 240 \text{ V}]$$

Final Answer

$$[E_g = 240 \text{ V}]$$

Question 2

Transformer Efficiency

Given

Rating = 3 kVA

OC Test

- 115 V
- 0.6 A
- 10 W

SC Test

- 60 V
- 13 A
- 200 W

Find efficiency at

- Full load
- Half load

at 0.8 pf lagging.

Step 1: Iron loss

OC test gives

[$P_i = 10\text{W}$]

Step 2: Copper loss

SC test gives full-load copper loss

[$P_{Cu} = 200\text{W}$]

Step 3: Output at Full Load

Transformer rating

[=3000VA]

Power factor

[=0.8]

Real output

[=3000×0.8 =2400W]

Step 4: Total Loss

[=10+200 =210W]

Step 5: Efficiency

[$\eta = (\text{Output} / (\text{Output} + \text{Loss}))$]

[= (2400 / (2400 + 210))]

[=0.9195]

[=91.95%]

Full-load Efficiency

[91.95%]

Half Load

Copper loss varies as

$$[I^2]$$

Half-load copper loss

$$[= (0.5)^2 \times 200]$$

$$[= 50W]$$

Output

$$[= 1500 \times 0.8 = 1200W]$$

Total loss

$$[= 10 + 50 = 60W]$$

Efficiency

$$[= (1200 / (1200 + 60))]$$

$$[= 95.24\%]$$

Final Answer

Half-load efficiency

$$[95.24\%]$$

Question 3

8-pole DC shunt generator

Wave winding

Given

- $P = 8$
- $Z = 778$
- $N = 500 \text{ rpm}$
- Load resistance = 12.5Ω
- Terminal voltage = $250V$
- Armature resistance = 0.24Ω
- Field resistance = 250Ω

Find

- Armature current
 - Induced EMF
 - Flux/pole
-

Step 1 Load Current

$$[I_L = (250 / (12.5)) = 20A]$$

Step 2 Field Current

$$[I_f = (250 / 250) = 1A]$$

Step 3 Armature Current

$$[I_a = I_L + I_f]$$

$$[= 20 + 1 = 21A]$$

Step 4 Generated EMF

For generator

$$[E_g = V + I_a R_a]$$

$$[= 250 + (21)(0.24)]$$

$$[= 255.04V]$$

Step 5 Flux per Pole

Wave winding

$$[A=2]$$

EMF equation

$$[255.04 = ((8\pi \phi (778)(500)) / (60 \times 2))]$$

$$[255.04 = ((3112000\pi \phi) / 120)]$$

$$[255.04 = 25933.33\pi \phi]$$

$$[\phi = 0.00984 \text{ Wb}]$$

Final Answers

Armature current

$$[21 \text{ A}]$$

Generated EMF

$$[255 \text{ V}]$$

Flux/pole

$$[0.00984 \text{ Wb}]$$

Question 4

Shunt generator

Given

Output current = 450A

Voltage = 230V

Field resistance = 50Ω

Armature resistance = 0.03Ω

Find generated EMF.

Step 1 Field Current

$$[I_f = (230/50) = 4.6A]$$

Step 2 Armature Current

$$[I_a = 450 + 4.6 = 454.6A]$$

Step 3 Generated EMF

$$[E_g = 230 + (454.6)(0.03)]$$

$$[= 243.64V]$$

Final Answer

$$[243.64V]$$

Question 5

4-pole generator

Given

- $P = 4$
- Speed = 1200 rpm
- Lap winding
- Slots = 65
- Conductors/slot = 12

- Flux = 0.02Wb
-

Step 1 Total Conductors

$$[Z = 65 \times 12 = 780]$$

Step 2 Parallel Paths

Lap winding

$$[A = P = 4]$$

Step 3 EMF Equation

$$[E = \frac{P \phi n Z}{60 A}]$$

$$[= \frac{(4 \times 0.02 \times 780 \times 1200)}{(60 \times 4)}]$$

Again, 4 cancels.

$$[= \frac{(0.02 \times 780 \times 1200)}{60}]$$

$$[= 312V]$$

Final Answer

$$[312V]$$

Let's solve it step by step.

Formulae to Memorize (3-Phase Induction Motor)

1. Synchronous Speed

$$[N_s = (120f/P)]$$

where:

- (N_s) = Synchronous speed (rpm)
 - (f) = Supply frequency (Hz)
 - (P) = Number of poles
-

2. Percentage Slip

$$[\%S = ((N_s - N)/(N_s)) \times 100]$$

where:

- (N) = Rotor (shaft) speed
-

3. Slip (Decimal)

$$[S = ((N_s - N)/(N_s))]$$

4. Rotor Speed

$$[N = N_s(1 - S)]$$

5. Slip Speed

$$[\text{Slip Speed} = N_s - N]$$

Given

- Number of poles,

[$P=6$]

- Frequency,

[$f=50$ Hz]

- Rotor speed,

[$N=960$ rpm]

Find

1. Percentage slip
 2. Slip speed
-

Step 1: Calculate Synchronous Speed

Formula

[$N_s=(120f/P)$]

Substitute values

[$N_s=((120 \times 50)/6)$]

[$N_s=(6000/6)$]

[$N_s=1000$ rpm]

Step 2: Calculate Slip Speed

Formula

[Slip Speed= N_s-N]

Substitute values

$$[= 1000 - 960]$$

$$[= 40 \text{ rpm}]$$

Step 3: Calculate Percentage Slip

Formula

$$[\%S = ((N_s - N) / (N_s)) \times 100]$$

Substitute values

$$[= ((1000 - 960) / 1000) \times 100]$$

$$[= (40 / 1000) \times 100]$$

$$[= 0.04 \times 100]$$

$$[= 4\%]$$

Final Answer

- **Synchronous Speed = 1000 rpm**
 - **Slip Speed = 40 rpm**
 - **Percentage Slip = 4%**
-
-

Let's solve it step by step. These questions are based on the formula for **electrical energy consumption** and **electricity bill calculation**.

Important Formulae to Memorize

1. Electrical Power

[P = Power in watts (W)]

Convert watts to kilowatts:

[$kW = (W/1000)$]

2. Energy Consumed

[$\text{Energy (kWh)} = \text{Power (kW)} \times \text{Time (hours)}$]

Remember:

- 1 kWh = 1 Unit of electricity
-

3. Total Energy

[$\text{Total Energy} = \text{Sum of energy consumed by all appliances}$]

4. Electricity Bill

[$\text{Energy Charges} = \text{Units} \times \text{Rate per Unit}$]

5. Total Bill

[$\text{Total Bill} = \text{Energy Charges} + \text{Fixed Charges}$]

Given

Fans

- Number = 5
- Power = 75 W each
- Time = 6 h/day

LED Batten Lights

- Number = 5
- Power = 22 W each
- Time = 6 h/day

Projector

- Number = 1
- Power = 330 W
- Time = 2 h/day

Electricity Rate

[Rs.;4.25/kWh]

Fixed Charges

[Rs.;200/month]

Assume

[30 days/month]

Step 1: Energy Consumed by Fans

Total Power

$$[5 \times 75 = 375\text{W}]$$

Convert into kW

$$[375/1000 = 0.375\text{kW}]$$

Daily Energy

$$[0.375 \times 6 = 2.25\text{kWh}]$$

Monthly Energy

$$[2.25 \times 30 = 67.5\text{kWh}]$$

Fans consume 67.5 units/month.

Step 2: Energy Consumed by LED Lights

Total Power

$$[5 \times 22 = 110\text{W}]$$

Convert to kW

$$[110/1000 = 0.11\text{kW}]$$

Daily Energy

$$[0.11 \times 6 = 0.66\text{kWh}]$$

Monthly Energy

$$[0.66 \times 30 = 19.8\text{kWh}]$$

LED lights consume 19.8 units/month.

Step 3: Energy Consumed by Projector

Power

$$[330\text{W} = 0.33\text{kW}]$$

Daily Energy

$$[0.33 \times 2 = 0.66 \text{kWh}]$$

Monthly Energy

$$[0.66 \times 30 = 19.8 \text{kWh}]$$

Projector consumes 19.8 units/month.

Step 4: Total Monthly Energy

$$[67.5 + 19.8 + 19.8 = 107.1 \text{kWh}]$$

Monthly energy consumption = 107.1 kWh (107.1 units).

Step 5: Energy Charges

$$[107.1 \times 4.25 = \text{Rs.} 455.175]$$

$\approx$

$$[\text{Rs.} 455.18]$$

Step 6: Add Fixed Charges

$$[455.18 + 200 = 655.18]$$

Final Answer

Monthly Energy Consumption

$$[107.1 \text{ kWh (units)}]$$

Energy Charges

[Rs.;455.18]

Total Electricity Bill

[Rs.;655.18]

Shortcut Method (Useful in Exams)

Appliance	Total Power (W)	Hours/day	Days	Energy (kWh)
5 Fans	375	6	30	67.5
5 LED Lights	110	6	30	19.8
Projector	330	2	30	19.8
Total	—	—	—	107.1

Bill:

- Energy Charges = $107.1 \times 4.25 = \text{Rs. } 455.18$
 - Fixed Charges = **Rs. 200**
 - **Total Bill = Rs. 655.18**
-

Exam Tips

1. Always convert **W** → **kW** by dividing by **1000**.
2. Calculate **daily energy** first, then multiply by the number of days for monthly energy.
3. **1 kWh = 1 unit** of electricity.
4. Don't forget to add **fixed charges** after calculating the energy charges.
5. Keep units correct:
 - Power → W or kW
 - Energy → kWh (units)

- Cost → Rupees (Rs.)

